MATH70108 Coursework

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Abstract

Fernando and Nerea Vadillo apply Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) to a multivariate time series of major equity indices, namely the DJI, FTSE, IBEX, CAC, DAX, and N225, with the aim of expressing some indices as linear functions of the others (Vadillo & Vadillo 2021). We rigorously analyse their methodology and extend their experiments, showing that the reported predictive accuracy is driven by experimental design rather than by stable cross-market structure. In particular, the odd-even split of training and test data introduces data leakage through time-series autocorrelation. When the data are split in a strictly forward-looking way, performance drops sharply. We also compute explicit low-rank reconstruction errors that the original paper omits. We further relate the poor reconstruction performance of some indices to the condition number of each problem. These results indicate that while PCA is useful for producing a low-rank linear approximation of the data, it cannot identify a stable relationship that can be relied on for forecasting across time.

1 Claims and Assumptions

Before diving into the main methodology introduced by the paper, we briefly highlight the implications of data processing and imputation.

The data are centred and normalised with respect to the maximum absolute deviation from the mean, restricting all values to $[-1, 1]$. The paper motivates this choice as an attempt to put the indices on a comparable scale, so that PCA focuses on the relative shape and co-movement of the time series, rather than being dominated by more volatile markets. However, this also means that the entire series is normalised relative to one extreme price level, so typical day-to-day movements are compressed and the resulting covariance reflects how indices align at their extremes rather than how they move over most of the year. This may bias the PCA toward identifying low-dimensional structure driven by shared extremes, which then overstates how well one index can be reconstructed from others across the full sample.

Forward-filling missing trading days also raises concerns about underestimating eigenvalues and overweighting open markets, since closed markets are forced to remain flat while others continue to move, which affects the covariance structure. This can artificially increase the perceived co-movement between indices, biasing the PCA toward finding stronger linear dependence between them. While this is not ideal, it mainly affects short-term movements rather than long-term market trends. We correct this behaviour in additional experiments by using the intersection calendar and restricting the analysis to days where all markets are open.

Eckart-Young-Mirsky Theorem A central mathematical motivation for the paper's procedure is the Eckart-Young-Mirsky (EYM) theorem covered during lectures. Let $A \in R^{m \times n}$ have SVD $A = U\Sigma V^\top$ with

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singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$ ($r = \operatorname{rank}(A)$). For $0 \leq k \leq r$, define the truncated SVD

$$A_k := U_k \Sigma_k V_k^\top = \sum_{i=1}^{k} \sigma_i \, u_i v_i^\top.$$

Then A_k is the *best* rank-k approximation to A in both spectral and Frobenius norms with optimal errors (setting $\sigma_{r+1} := 0$):

$$\min_{\operatorname{rank}(B)\leq k} \|A - B\|_2 = \sigma_{k+1}, \qquad \min_{\operatorname{rank}(B)\leq k} \|A - B\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2.$$

According to the Eckart-Young-Mirsky theorem, truncating the SVD provides an optimal low-dimensional reconstruction (Eckart & Young 1936, Golub & Van Loan 2013). This is the foundation behind the paper's k-dimensional approximation. While EYM guarantees that the truncated SVD gives the best rank-k approximation to the data matrix, the paper goes further by treating this approximation as a structural model, implicitly assuming that the market index vectors lie close to a fixed low-dimensional linear subspace.

Mapping the paper's experiment to EYM Let $P \in R^{254\times 6}$ denote the cleaned data matrix used in the paper. Write the train/test split as P_1 and P_2, and let p_i^{tr}, $p_j^{\text{te}} \in R^{1\times 6}$ denote generic rows of P_1 and P_2, respectively. On the training set P_1, the paper computes an SVD

$$P_1 = U\Sigma V^\top, \qquad V = [v_1, \ldots, v_6], \quad v_\ell \in R^6,$$

and forms k-dimensional reconstructions by restricting to the span of the first k right singular vectors. For a given k, we write $\hat{p}_i^{(k)}$ for the rank-k *reconstruction* of a row p_i^{tr}:

$$\hat{p}_i^{(k)} \in \operatorname{span}\{v_1^\top, \ldots, v_k^\top\},$$

At this point the paper implicitly assumes that the true market vectors lie close to this k-dimensional subspace, so that the truncated SVD is not merely a compression of the training data but a structural low-factor model of the six indices. Under this assumption, each market vector admits a rank-k approximation

$$\hat{p}_i^{(k)} = \sum_{\ell=1}^{k} \alpha_{i\ell} v_\ell^\top, \qquad k = 1, \ldots, 5,$$

where the coefficients $\alpha_{i\ell}$ are the coordinates of $\hat{p}_i^{(k)}$ in the basis $\{v_1, \ldots, v_k\}$. Since V is orthogonal, the least-squares choice is the orthogonal projection onto $\operatorname{span}\{v_1, \ldots, v_k\}$:

$$\hat{p}_i^{(k)} = p_i^{\text{tr}} V_k V_k^\top, \qquad \alpha_i := p_i^{\text{tr}} V_k \in R^{1\times k},$$

so that in particular $\alpha_{i\ell} = p_i^{\text{tr}} v_\ell$.

Test-set reconstruction in the paper Fix V_k learned from P_1. In the paper's k-dimensional prediction experiments, one assumes that, for each test day j, only a subset of coordinates $S \subset \{1, \ldots, 6\}$ with $|S| = k$ is observed. Writing $p_{j,S}^{\text{te}} \in R^{1\times k}$ for the observed entries and $(V_k)_S \in R^{k\times k}$ for the submatrix of V_k formed by selecting the rows indexed by S, the coefficients $\beta_j \in R^{1\times k}$ are obtained by solving the $k \times k$ linear system

$$p_{j,S}^{\text{te}} \approx \beta_j \big((V_k)_S\big)^\top, \qquad \text{i.e.} \qquad \beta_j \approx p_{j,S}^{\text{te}} \left(\big((V_k)_S\big)^\top\right)^{-1},$$

(when invertible), and the full 6-dimensional reconstruction is then

$$\widehat{p}_j^{(k)} := \beta_j V_k^\top.$$

Thus, for $k = 2$ the method reduces to a 2×2 system per day, and for $k = 5$ to a 5×5 system per day. Clearly, the stability of the prediction procedure is directly tied to the numerical stability of inverting $\big((V_k)_S\big)^\top$, which is heavily influenced by how the data are imputed or normalised.

Reconstruction error and fitness implications (in-sample vs. out-of-sample) Define the global reconstruction coefficient of determination for a general (not necessarily centred) data matrix X by

$$R_k^2 := 1 - \frac{\mathrm{SSE}_k}{\mathrm{TSS}}, \qquad \mathrm{SSE}_k := \|X - \hat{X}_k\|_F^2, \quad \mathrm{TSS} := \|X - \bar{X}\|_F^2,$$

where $\bar{X}$ is the matrix obtained by repeating the column means of X across rows. When X is centred, $\bar{X} = 0$. In this case, the rank-k truncated SVD X_k is the unique matrix that minimises the Frobenius norm error $\|X - Y\|_F^2$ over all matrices Y of rank at most k. This yields the standard explained-variance formula, which is consistent with the relative error formula quoted by the paper (Jolliffe & Cadima 2016)

$$R_k^2 = \frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2} \in [0, 1].$$

However, the paper applies the training subspace V_k to test out-of-sample observations $p_i^{\mathrm{te}} \in P_2$ via $\hat{p}_i^{(k)} = p_i^{\mathrm{te}} V_k V_k^\top$, where this projection is not the rank-k minimiser for P_2. Thus there is no guarantee that $\mathrm{SSE}_k \leq \mathrm{TSS}$ on the test set, so test-set R_k^2 can be negative. This distinction is central to our critique: the method is motivated by EYM in-sample optimality, but the paper does not report an explicit out-of-sample fitness metric. The paper implicitly carries the in-sample intuition into an out-of-sample task without measuring fit, which is a critical error in analysis.

2 Methodology

To demonstrate the flaws of the original paper that we analysed in the previous sections, we set up the following experiments. We introduce three control groups A, B, and C, which correspond to a faithful replication of the original paper, an experiment after correcting data leakage, and an experiment performed on the intersection calendar. We repeat the $k = 1, 2, 5$ reconstruction scenarios in the original paper across all control groups. We collected six market indices (DJI, NI225, DAX, CAC, IBEX, and FTSE) over the year 2018 via Yahoo Finance and compiled data for 254 days based on the EU calendar, like the original paper. All of the experiments are implemented using sklearn and numpy (Suresh et al. 2026).

Perhaps the most critical error in the original paper is in its evaluation of reconstruction, which is phrased as a predictive analysis. Although this is technically an 'out-of-sample' analysis, it is a form of data leakage, since the test set was taken from a series directly correlated with the training set. Control group B corrects this protocol by splitting the data such that the training set consists of all data up to August 31, 2018, and the test set consists of data from August 31, 2018 onwards. Control group C is equivalent to group A, except that it performs an analysis on the intersection calendar (only on days when all markets were open), rather than forward filling null values.

We report and analyse the global reconstruction coefficient of determination discussed in the previous section, in order to analyse how the reconstruction quality varies between the 3 control groups. We also report the condition numbers to observe the sensitivity of the procedure. Based on our work on matrix inversion and error analysis, we hypothesise that the structure revealed by the original paper is primarily an artefact of the ill-posed experimental set-up. Therefore, we predict that the result will vary between all 3 control groups.

3 Results and Evaluation

We now focus on reporting quantitative findings omitted in the original paper and compare them with additional experiments. We succeeded in reproducing high-level trends identified by the original paper; plots and metrics that are not relevant to the main analysis are included in the supplement.

One- and two-dimensional reconstruction In Table 1a, the R^2 values of control group A appear to lie between 0 and 1, suggesting some level of explanatory power. By construction, the observed indices

Table 1: Paper-style reconstruction performance (R^2 on the test set) across control groups A–C.

(a) 1D reconstruction ($k = 1$, observed CAC only).

Index	A	B	C
CAC	1.000	1.000	1.000
DAX	0.864	0.694	0.878
DJI	0.221	-0.820	0.215
FTSE	0.856	0.152	0.856
IBEX	0.663	-0.439	0.675
N225	0.510	-1.897	0.493

(b) 2D reconstruction ($k = 2$, observed CAC and DAX).

Index	A	B	C
CAC	1.000	1.000	1.000
DAX	1.000	1.000	1.000
DJI	-1.119	-1.909	-1.112
FTSE	0.863	-0.039	0.857
IBEX	0.923	0.713	0.924
N225	-0.678	-2.934	-0.499

Table 2: 5D leave-one-out reconstruction: performance and stability diagnostics.

(a) 5D leave-one-out R^2 (test) by held-out index ($k = 5$).

Held-out	A	B	C
CAC	0.858	-1.511	0.893
DAX	0.975	0.824	0.974
DJI	-50.681	-44.218	-497.161
FTSE	-1.994	-7.359	-12.564
IBEX	0.919	-0.170	0.910
N225	-13.125	-36.355	-32.104

(b) Condition numbers κ of the 5D held-out reconstruction systems.

Held-out index	A	B	C
CAC	2.558	5.833	2.208
DAX	1.227	1.209	1.251
DJI	38.534	21.448	146.921
FTSE	14.666	5.643	31.630
IBEX	2.380	2.000	2.549
N225	28.943	17.740	44.144

achieve $R^2 = 1$ across all control groups. However, this impression is misleading. When the odd–even split is replaced by a temporally consistent split in control group B, reconstruction quality deteriorates sharply, with several indices exhibiting negative R^2 values. In particular, DJI, IBEX, and N225 fall below zero in the one-dimensional case. Negative R^2 indicates that the reconstruction performs worse than the trivial baseline of simply predicting the sample mean, clearly contradicting the claims of predictive structure.

Table 1b further reinforces this conclusion. Even under control group A, where data leakage remains present, reconstruction quality degrades for several indices, with DJI and N225 exhibiting strongly negative R^2. Under control group B, the deterioration becomes more pronounced, indicating that the apparent stability of low-dimensional structure is highly sensitive to the experimental design. Control group C partially recovers the qualitative behaviour observed in group A, but still exhibits substantial instability across indices.

Taken together, these results demonstrate that the bounded $R^2 \in [0, 1]$ values observed in the original paper are not due to the merits of the method, but are instead the result of the chosen data split and the unorthodox pre-processing approaches (forward filling and global range normalisation).

Five-dimensional reconstruction This setting is particularly informative, as it corresponds to the highest-rank approximation considered in the original paper while still requiring genuine extrapolation.

In Table 2a, we observe severe failures across all control groups. Reconstructions of DJI and N225 yield strongly negative R^2 values across all control groups, indicating a systematic breakdown of the assumed low-rank structure. These failures persist and sometimes worsen under control groups B and C, confirming that they are not just results of the flawed data split but instead reflect fundamental instability in the reconstruction methodology.

Numerical stability Table 2b reports condition numbers κ of the linear systems solved in the five-dimensional reconstructions. The condition number is computed as the ratio of the largest to the smallest singular value of the matrix obtained by restricting $V_k^\top$ to the observed coordinates. Large condition numbers

indicate sensitivity to small perturbations in the data and thus poor numerical stability.

A clear pattern emerges. According to Table 2b, reconstructions involving CAC, DAX, and IBEX are relatively well-conditioned across all control groups, with condition numbers typically in the range 1–5. In contrast, reconstructions involving FTSE, DJI, and N225 exhibit condition numbers one or two orders of magnitude larger than those for other indices. This behaviour directly explains the erratic reconstruction performance observed in Table 2a: ill-conditioned systems amplify noise and estimation error, leading to highly unstable reconstructions and large negative R^2 values.

The original paper characterises the poor FTSE performance as a "surprise." Our results indicate that such behaviour is entirely expected given the set-up of the underlying reconstruction problem. The instability is not indicative of anomalous market behaviour, but rather a direct consequence of the linear algebra induced by the chosen reconstruction scheme.

4 Conclusion

In this report, we explicitly mapped the procedure followed by Fernando and Nerea Vadillo to the Eckart-Young-Mirsky (EYM) theorem, enabling us to identify the key implicit assumption underlying their analysis, which is that the low-dimensional subspace learned from the training data represents a stable structural relationship that generalises out-of-sample. While EYM guarantees that the truncated SVD provides an optimal rank-k approximation in-sample, it offers no guarantee of predictive accuracy when the learned subspace is applied to unseen data. The original paper implicitly extends this in-sample optimality to an out-of-sample reconstruction task without reporting explicit test-set reconstruction errors. Our analysis shows that this extension is not justified. When reconstruction quality is measured using an explicit out-of-sample R^2, performance can deteriorate sharply. Moreover, our analysis also shows that the reported predictive patterns are driven by experimental design choices rather than by a stable cross-market structure.

At the same time, our results do not contradict the weaker and more limited claim that equity indices exhibit synchronous co-movement. Low-rank approximations indeed capture shared variance in index levels, which is unsurprising given common macroeconomic influences. However, this observation alone does not support the stronger interpretation of PCA as a reliable predictive model across time. Our explicit reconstruction errors and the frequent occurrence of negative out-of-sample R^2 demonstrate that the inferred low-dimensional structure does not generalise in a stable way.

Our study can be further extended by tackling various limitations in the present work. First, the analysis treats daily index levels as synchronous observations, despite the fact that global markets operate in different time zones. In practice, trading follows a clear temporal order with only limited overlap between European and US market hours. As a result, the only economically meaningful notion of prediction across markets would involve forecasting later-opening markets from earlier ones. Second, the analysis focuses on variance structure in normalised index levels. A more standard approach in quantitative finance would work with log returns, which reduce nonstationarity, remove much of the autocorrelation present in price levels, and better align with economic interpretations of risk and dependence. Extending the analysis to returns and combining PCA with time-series models designed for forecasting, would provide a more appropriate framework for assessing whether low-dimensional structure can support genuine predictive content.

Overall, our findings indicate that while PCA is a powerful tool for dimensionality reduction and descriptive analysis, it does not, in this setting, identify a stable linear relationship that can be relied upon for out-of-sample prediction. The predictive patterns reported in the original paper are best understood as artefacts of experimental design rather than evidence of robust cross-market structure.

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